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SQL Fundamentals : SELECT and Filtering

Exercise 1 : SQL Foundations

employees_table

id	first_name	last_name	department	salary	hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

1. Retrieve all columns from the employees table

```
SELECT *
FROM employees_table;
```

results

id	first_name	last_name	department	Salary	hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. Find all unique departments
EXPECTED COLUMNS: departments

```
SELECT DISTINCT departments
FROM employees_table;
```

outcome →

results_table

departments
IT
HR
Finance
Marketing

3. Retrieve first and last names ordered by salary descending
EXPECTED COLUMNS: first_name last_name

```
SELECT first_name,
       last_name
FROM employees_table
ORDER BY salary DESC;
```

outcome →

results_table

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees.
EXPECTED COLUMNS: id, first_name, last_name, salary.

```
SELECT id,
       first_name,
       last_name,
       salary
FROM employees_table
ORDER BY salary DESC
LIMIT 3;
```

results_table

id	first_name	last_name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

5. Find employees in the IT department

EXPECTED COLUMNS : id, first_name, last_name, department

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees_table
WHERE department = 'IT';
```

results_table

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. Find employees in Finance with salary > 6000

EXPECTED COLUMNS : id, first_name, last_name, department, salary

```
SELECT id,
       first_name
       last_name,
       department,
       salary
FROM employees_table
WHERE department = 'FINANCE' AND salary > 60000;
```

results_table

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

7. Find employees in HR or Marketing

EXPECTED COLUMN : id, first_name, last_name, department

```
SELECT id,
       first-name,
       last_name,
       department
FROM employees_table
WHERE department = 'HR'
OR department = 'Marketing';
```

results_table

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. Find employees not in IT.
EXPECTED COLUMNS: id, first_name, last_name, department

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees_table
WHERE NOT department = 'IT';
```

results_table

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Find employees in IT, HR, or Finance using IN.
EXPECTED COLUMNS: id, first_name, last_name, department

```
SELECT id,
       first_name,
       last_name,
       department
FROM employees_table
WHERE department IN ('IT', 'HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 65000 and city Johannesburg
EXPECTED COLUMNS: id, first_name, last_name, department, salary, city

```
SELECT id,
       first_name,
       last_name,
       department,
       Salary,
       City
FROM employees_table
WHERE department = 'IT' AND Salary > 65000 AND city = 'Johannesburg';
```

results_table

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg